

Startup Folder Persistence Detection and Investigation using Sysmon and Wazuh

Objective

The objective of this simulation was to investigate how Sysmon and Wazuh detect Startup Folder persistence techniques commonly associated with malware persistence, unauthorized access retention, and post-compromise attacker activity.

MITRE ATTACK Mapping

Tactic	Technique	ID
Persistence	Startup Folder	T1547.001
Execution	PowerShell	T1059.001

Attack Simulation

The following command was executed in the Windows endpoint lab environment:

```
powershell.exe -Command "New-Item 'C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\evil.txt' -ItemType File"
```

Attack Overview

Step	Activity
1	PowerShell launched
2	File created inside Startup folder
3	Sysmon captured persistence artifact creation
4	Wazuh generated persistence detection alert
5	User Logon
6	Payload Auto-Executed

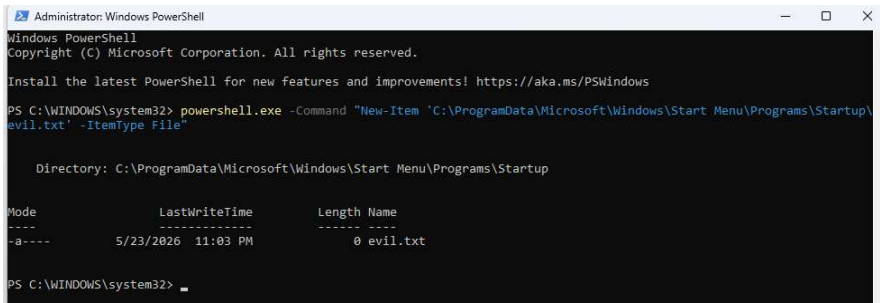


Figure 1 — Startup folder persistence simulation executed in the lab environment

Persistence Investigation

Key Sysmon Events Identified

Sysmon Event ID	Detection Purpose
1	Process Creation
11	File Creation

These events formed the primary telemetry chain used during the persistence investigation.

Sysmon Event ID 1 – PowerShell Process Creation Analysis

Key Findings

Field	Value
Event ID	1
Process	powershell.exe
Parent Process	powershell.exe
User	socadmin
Integrity Level	High

Investigation Analysis

Sysmon Event ID 1 captured PowerShell execution responsible for creating a persistence artifact inside the Windows Startup folder.

PowerShell involvement significantly increased suspicion because attackers frequently abuse PowerShell to establish persistence mechanisms while avoiding traditional executable-based detections.

The nested PowerShell parent-child relationship further strengthened behavioral suspicion.

Operational Number of events: 10,231					
Level	Date and Time	Source	Event ID	Task C...	
Information	5/23/2026 11:03:52 PM	Sysmon	11	File cre...	
Information	5/23/2026 11:03:48 PM	Sysmon	11	File cre...	
Information	5/23/2026 11:03:47 PM	Sysmon	1	Proces...	
Information	5/23/2026 11:02:00 PM	Sysmon	11	File cre...	
Information	5/23/2026 11:01:59 PM	Sysmon	1	Proces...	
Information	5/23/2026 11:01:57 PM	Sysmon	1	Proces...	
Information	5/23/2026 11:01:57 PM	Sysmon	1	Proces...	
Event 1, Sysmon					
General Details					
ProcessId: 3584 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe FileVersion: 10.0.26100.5074 (WinBuild.160101.0800) Description: Windows PowerShell Product: Microsoft® Windows® Operating System Company: Microsoft Corporation OriginalFileName: PowerShell.EXE CommandLine: "C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -Command "New-Item 'C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\evil.txt' -ItemType File" CurrentDirectory: C:\WINDOWS\system32\ User: DESKTOP-HDF09SQ\socadmin LogonGuid: {842f9a06-b3f2-6a11-840c-0a0000000000} LogonId: 0xA0C84 TerminalSessionId: 1 IntegrityLevel: High Hashes: MD5=A97E6573B97B44C96122BFA543A82EA1,SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5F56381870317E8C46,IMPHASH=AFACF6DC9041114B198160AAB4D9A8F77 Log Name: Microsoft-Windows-Sysmon/Operational Source: Sysmon Logged: 5/23/2026 11:03:47 PM Event ID: 1 Task Category: Process Create (rule: ProcessCreate) Level: Information Keywords: User: SYSTEM Computer: DESKTOP-HDF09SQ OpCode: Info					

Figure 2 — Sysmon Event ID 1 showing PowerShell execution responsible for persistence creation

Sysmon Event ID 11 – Startup Folder File Creation Analysis

Key Findings

Field	Value
Event ID	11
Source Process	powershell.exe
Target Filename	Startup\evil.txt
User	socadmin

Critical Persistence Artifact

TargetFilename

```
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\evil.txt
```

This was the primary persistence artifact identified during the investigation.

Investigation Analysis

Sysmon Event ID 11 captured the creation of a file within the Windows Startup folder by PowerShell.

The Startup folder is a high-value persistence location because files or shortcuts placed within this directory may automatically execute during user logon and session initialization.

Although the file created during testing was a benign `.txt` file, the behavioral pattern itself remained highly suspicious because PowerShell was used to create a persistence-related artifact.

In real-world attacks, adversaries commonly place:

- `.lnk`
- `.bat`
- `.ps1`
- `.vbs`
- `.exe`

files within Startup folders to maintain persistence after system reboot or user login.

Important SOC Analyst Note

MITRE ATT&CK mappings associated with telemetry sources are not always perfectly accurate.

The Sysmon rule originally referenced legacy persistence mappings, while the modern ATT&CK technique more accurately aligns with:

Technique	ID
Startup Folder Persistence	T1547.001

This highlights the importance of analyst validation rather than relying entirely on automated SIEM or telemetry mappings.

Operational Number of events: 10,231				
Level	Date and Time	Source	Event ID	Task C...
Information	5/23/2026 11:03:52 PM	Sysmon	11	File cre...
Information	5/23/2026 11:03:48 PM	Sysmon	11	File cre...
Information	5/23/2026 11:03:47 PM	Sysmon	1	Proces...
Information	5/23/2026 11:02:00 PM	Sysmon	11	File cre...
Information	5/23/2026 11:01:59 PM	Sysmon	1	Proces...
Information	5/23/2026 11:01:57 PM	Sysmon	1	Proces...
Information	5/23/2026 11:01:57 PM	Sysmon	1	Proces...

Event 11, Sysmon	
General	Details
File created: RuleName: T1023 UtcTime: 2026-05-23 17:33:52.232 ProcessGuid: {842f9a06-e4fb-6a11-cc02-000000001c00} ProcessId: 3584 Image: C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\evil.txt CreationUtcTime: 2026-05-23 17:33:52.229 User: DESKTOP-HDF09SQ\socadmin	
Log Name:	Microsoft-Windows-Sysmon/Operational
Source:	Sysmon
Event ID:	11
Level:	Information
User:	SYSTEM
OpCode:	Info
Logged:	5/23/2026 11:03:52 PM
Task Category:	File created (rule: FileCreate)
Keywords:	
Computer:	DESKTOP-HDF09SQ

Figure 3 — Sysmon Event ID 11 showing Startup folder persistence artifact creation

Persistence Severity Assessment

Investigation Findings

Activity	Severity
Generic PowerShell execution	Medium
Encoded PowerShell	High
Startup persistence creation	Very High

Investigation Analysis

Persistence activity significantly increases incident severity because it indicates potential attacker intent to maintain long-term access within the environment.

If analysts observed the following process chain in an enterprise environment:

```
powershell.exe
↓
creates file in Startup folder
```

they would immediately investigate:

- malware persistence
- user compromise
- attacker staging activity

- LOLBin abuse
- ransomware preparation

This type of behavior is not considered normal administrative activity.

Persistence Execution After Logon

Sysmon Event ID 1 – Process Creation

Key Findings

Field	Value
Event ID	1
Process	Notepad.exe
Parent Process	explorer.exe
User	socadmin
Integrity Level	Medium
Startup File	evil.txt

Investigation Analysis

Following user logon after system restart, Sysmon Event ID 1 recorded execution of the Startup Folder persistence artifact.

The event shows:

Parent Process: explorer.exe

launching: Notepad.exe

with:

C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\evil.txt

as the argument.

This confirms the Startup Folder persistence mechanism successfully executed automatically after user authentication.

Unlike file creation events, this telemetry provides direct evidence that the persistence technique was triggered and executed during logon.

The event demonstrates:

- Persistence activation
- User logon execution
- Explorer-based startup processing
- Successful payload execution

MITRE ATT&CK:

T1547.001 - Registry Run Keys / Startup Folder

Operational

Number of events: 2,944 (!) New events available

Level	Date and Time	Source	Event ID	Task C...
Information	6/6/2026 4:44:48 PM	Sysmon	1	Proces...

Event 1, Sysmon

General

Details

Process Create:

RuleName: -

UtcTime: 2026-06-06 11:14:48.408

ProcessGuid: {842f9a06-0128-6a24-4201-000000003000}

ProcessId: 5948

Image: C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad\Notepad.exe

FileVersion: -

Description: -

Product: -

Company: -

OriginalFileName: -

CommandLine: "C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2604.5.0_x64__8wekyb3d8bbwe\Notepad\Notepad.exe" "C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\evil.txt"

CurrentDirectory: C:\Users\socadmin\

User: DESKTOP-HDF09SQ\socadmin

LogonGuid: {842f9a06-00d2-6a24-b8a4-0f0000000000}

LogonId: 0xFA4B8

TerminalSessionId: 1

IntegrityLevel: Medium

Hashes: MD5=28DB956AD9A153C763983BF39ED067F6,SHA256=3854252D7A10DB9E4E908012BDE027B89B42F9DF41B36886A9F66FE89398F552,IMPHASH=58431BC3B7BCC27A4223E8B554B8801F

ParentProcessGuid: {842f9a06-00d3-6a24-f700-000000003000}

ParentProcessId: 3500

ParentImage: C:\Windows\explorer.exe

ParentCommandLine: C:\WINDOWS\Explorer.EXE

ParentUser: DESKTOP-HDF09SQ\socadmin

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon

Logged: 6/6/2026 4:44:48 PM

Event ID: 1

Task Category: Process Create (rule: ProcessCreate)

Level: Information

Keywords:

User: SYSTEM

Computer: DESKTOP-HDF09SQ

OpCode: Info

Figure 5: Sysmon Event ID 1 showing automatic execution of Startup Folder persistence payload after user login.

Wazuh Detection Analysis

Wazuh Rules Triggered

Rule ID	Severity	Description
92027	4	PowerShell process spawned PowerShell instance
92213	15	Executable file dropped in folder commonly used by malware
554	5	File added to the system (FIM/syscheck)

Investigation Analysis

Wazuh successfully identified suspicious persistence-related activity through Sysmon file creation telemetry originating from PowerShell.

The detection logic correctly identified:

- PowerShell execution
- Startup folder modification
- persistence artifact creation

- suspicious file placement behavior

The combination of PowerShell activity and Startup folder modification significantly increased investigation confidence.

False Positive Validation

Although the created file was a benign `.txt` file used for testing, the behavioral pattern itself remained highly suspicious because attackers commonly use the same persistence location for malicious payload execution.

This highlights the importance of behavioral analysis and contextual investigation rather than relying solely on file reputation or extension-based detection.

```
> May 23, 2026 @ 23:03:58.778 syscheck.mode: realtime syscheck.path: c:\programdata\microsoft\windows\start menu\programs\startup\levil.txt
syscheck.sha1_after: da39a3ee5e6b4b6d3255bfe9961890afd80709 syscheck.uname_after: Administrators syscheck.mtime_after: May 23,
2026 @ 23:03:52.000 syscheck.attrs_after: ARCHIVE, NOT_CONTENT_INDEXED syscheck.size_after: 0 syscheck.uid_after: S-1-5-32-544
syscheck.win_perm_after.allowed: DELETE, DELETE, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA,
APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONI

> May 23, 2026 @ 23:03:56.551 input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.image: C:\\WINDOWS\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {842f9a06-
e4fb-6a11-cc02-000000001c00} data.win.eventdata.processId: 3584 data.win.eventdata.uptime: 2026-05-23 17:33:48.310
data.win.eventdata.targetFileName: C:\\Users\\socadmin\\AppData\\Local\\Temp\\_PSScriptPolicyTest_tb3b5ijp.d0q.ps1
data.win.eventdata.creationUtcTime: 2026-05-23 17:33:48.308 data.win.eventdata.user: DESKTOP-HDF09SQ\\socadmin

> May 23, 2026 @ 23:03:56.583 input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.originalFileName: PowerShell.EXE data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\pove
rshell.exe data.win.eventdata.product: Microsoft Windows Operating System data.win.eventdata.parentProcessGuid: {842f9a06-e48f-6
a11-ca02-000000001c00} data.win.eventdata.description: Windows PowerShell data.win.eventdata.logonGuid: {842f9a06-b3f2-6a11-840c-0
a0000000000} data.win.eventdata.parentCommandLine: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\"
```

Figure 4 – Wazuh detection for Startup folder persistence activity

```
{
  "_index": "wazuh-alerts-4.x-2026.05.23",
  "_id": "NxrmVZ4Bdl3fDzgsm7r",
  "_version": 1,
  "_score": null,
  "_source": {
    "input": {
      "type": "log"
    },
    "agent": {
      "ip": "192.168.56.106",
      "name": "DESKTOP-HDF09SQ",
      "id": "001"
    },
    "manager": {
      "name": "wazuh-server"
    },
    "data": {
      "win": {
        "eventdata": {
          "originalFileName": "PowerShell.EXE",
          "image": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
          "product": "Microsoft Windows Operating System",
          "parentProcessGuid": "{842f9a06-e48f-6a11-ca02-000000001c00}",
          "description": "Windows PowerShell",
          "logonGuid": "{842f9a06-b3f2-6a11-840c-0a0000000000}",
          "parentCommandLine": "\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\"",
          "processGuid": "{842f9a06-e4fb-6a11-cc02-000000001c00}",
          "logonId": "0xa0c84",
          "parentProcessId": "10508",
          "processId": "3584",
          "currentDirectory": "C:\\\\WINDOWS\\\\system32\\\\",
          "utcTime": "2026-05-23 17:33:47.227",
          "hashes":
            "MD5=A97E6573B97B44C96122BFA543A82EA1,SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5",
          "parentImage": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
```

```

"company": "Microsoft Corporation",
"commandLine": "\"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\"" -Command
\\\\"New-Item 'C:\\\\ProgramData\\\\Microsoft\\\\Windows\\\\Start Menu\\\\Programs\\\\Startup\\\\evil.txt' -ItemType
File\\\\",
"integrityLevel": "High",
"fileVersion": "10.0.26100.5074 (WinBuild.160101.0800)",
"user": "DESKTOP-HDF09SQ\\\\socadmin",
"terminalSessionId": "1",
"parentUser": "DESKTOP-HDF09SQ\\\\socadmin"
},
"system": {
"eventID": "1",
"keywords": "0x8000000000000000",
"providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
"level": "4",
"channel": "Microsoft-Windows-Sysmon/Operational",
"opcode": "0",
"message": "\"Process Create:\\\\r\\nRuleName: -\\r\\nUtcTime: 2026-05-23 17:33:47.227\\r\\nProcessGuid: {842f9a06-
e4fb-6a11-cc02-000000001c00}\\r\\nProcessId: 3584\\r\\nImage:
C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\r\\nFileVersion: 10.0.26100.5074
(WinBuild.160101.0800)\\r\\nDescription: Windows PowerShell\\r\\nProduct: Microsoft® Windows® Operating
System\\r\\nCompany: Microsoft Corporation\\r\\nOriginalFileName: PowerShell.EXE\\r\\nCommandLine:
'\\\\C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\" -Command \\\\'New-Item
' C:\\\\ProgramData\\\\Microsoft\\\\Windows\\\\Start Menu\\\\Programs\\\\Startup\\\\evil.txt' -ItemType File\\\\'\\r\\nCurrentDirectory:
C:\\\\WINDOWS\\\\system32\\\\\\r\\nUser: DESKTOP-HDF09SQ\\\\socadmin\\r\\nLogonGuid: {842f9a06-b3f2-6a11-840c-
0a0000000000}\\r\\nLogonId: 0xA0C84\\r\\nTerminalSessionId: 1\\r\\nIntegrityLevel: High\\r\\nHashes:
MD5=A97E6573B97B44C96122BFA543A82EA1,SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5F
{842f9a06-e48f-6a11-ca02-000000001c00}\\r\\nParentProcessId: 10508\\r\\nParentImage:
C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\r\\nParentCommandLine:
'\\\\C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\" \\r\\nParentUser: DESKTOP-
HDF09SQ\\\\socadmin\\\\',
"version": "5",
"systemTime": "2026-05-23T17:33:47.2746323Z",
"eventRecordID": "15379",
"threadID": "4376",
"computer": "DESKTOP-HDF09SQ",
"task": "1",
"processID": "3596",
"severityValue": "INFORMATION",
"providerName": "Microsoft-Windows-Sysmon"
}
},
"rule": {
"firedtimes": 7,
"mail": false,
"level": 4,
"description": "Powershell process spawned powershell instance",
"groups": [
"sysmon",
"sysmon_eid1_detections",
"windows"
],
"mitre": {
"technique": [
"PowerShell"
],
"id": [
"T1059.001"
]
}
}

```



```

    "tactic": [
      "Execution"
    ],
    "id": "92027",
    "location": "EventChannel",
    "decoder": {
      "name": "windows_eventchannel"
    },
    "id": "1779557636.3767710",
    "timestamp": "2026-05-23T17:33:56.503+0000",
    "fields": {
      "timestamp": [
        "2026-05-23T17:33:56.503Z"
      ]
    },
    "sort": [
      1779557636503
    ]
  }

```

Wazuh telemetry showing PowerShell parent-child process behavior associated with persistence activity

```

{
  "_index": "wazuh-alerts-4.x-2026.05.23",
  "_id": "ORrmVZ4Bdld3fDzgsm7r",
  "_version": 1,
  "_score": null,
  "_source": {
    "syscheck": {
      "mode": "realtime",
      "path": "c:\\programdata\\microsoft\\windows\\start menu\\programs\\startup\\levil.txt",
      "sha1_after": "da39a3ee5e6b4b0d3255bfef95601890afd80709",
      "uname_after": "Administrators",
      "mtime_after": "2026-05-23T17:33:52",
      "attrs_after": [
        "ARCHIVE",
        "NOT_CONTENT_INDEXED"
      ],
      "size_after": "0",
      "uid_after": "S-1-5-32-544",
      "win_perm_after": [
        {
          "allowed": [
            "DELETE",
            "READ_CONTROL",
            "WRITE_DAC",
            "WRITE_OWNER",
            "SYNCHRONIZE",
            "READ_DATA",
            "WRITE_DATA",
            "APPEND_DATA",
            "READ_EA",
            "WRITE_EA",
            "EXECUTE",
            "READ_ATTRIBUTES",
            "WRITE_ATTRIBUTES"
          ]
        }
      ]
    }
  }
}

```

```

"DELETE",
"READ_CONTROL",
"WRITE_DAC",
"WRITE_OWNER",
"SYNCHRONIZE",
"READ_DATA",
"WRITE_DATA",
"APPEND_DATA",
"READ_EA",
"WRITE_EA",
"EXECUTE",
"READ_ATTRIBUTES",
"WRITE_ATTRIBUTES"
],
[
"READ_CONTROL",
"SYNCHRONIZE",
"READ_DATA",
"READ_EA",
"EXECUTE",
"READ_ATTRIBUTES"
],
[
"READ_CONTROL",
"SYNCHRONIZE",
"READ_DATA",
"READ_EA",
"EXECUTE",
"READ_ATTRIBUTES"
]
],
"name": "S-1-5-21-221611468-1794481200-2977160085-1000"
},
{
"allowed": [
"DELETE"
],
"name": "socadmin"
},
{
"allowed": [
"DELETE",
"READ_CONTROL",
"WRITE_DAC",
"WRITE_OWNER",
"SYNCHRONIZE",
"READ_DATA",
"WRITE_DATA",
"APPEND_DATA",
"READ_EA",
"WRITE_EA",
"EXECUTE",
"READ_ATTRIBUTES",
"WRITE_ATTRIBUTES"
],
"name": "SYSTEM"
},

```

```

{
  "allowed": [
    "DELETE",
    "READ_CONTROL",
    "WRITE_DAC",
    "WRITE_OWNER",
    "SYNCHRONIZE",
    "READ_DATA",
    "WRITE_DATA",
    "APPEND_DATA",
    "READ_EA",
    "WRITE_EA",
    "EXECUTE",
    "READ_ATTRIBUTES",
    "WRITE_ATTRIBUTES"
  ],
  "name": "Administrators"
},
{
  "allowed": [
    "READ_CONTROL",
    "SYNCHRONIZE",
    "READ_DATA",
    "READ_EA",
    "EXECUTE",
    "READ_ATTRIBUTES"
  ],
  "name": "Users"
},
{
  "allowed": [
    "READ_CONTROL",
    "SYNCHRONIZE",
    "READ_DATA",
    "READ_EA",
    "EXECUTE",
    "READ_ATTRIBUTES"
  ],
  "name": "Everyone"
}
],
"event": "added",
"md5_after": "d41d8cd98f00b204e9800998ecf8427e",
"sha256_after": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855"
},
"input": {
  "type": "log"
},
"agent": {
  "ip": "192.168.56.106",
  "name": "DESKTOP-HDF09SQ",
  "id": "001"
},
"manager": {
  "name": "wazuh-server"
},
"rule": {
  "firedtimes": 2,
  "mail": false,
  "level": 5,

```

```

"pci_dss": [
  "11.5"
],
"hipaa": [
  "164.312.c.1",
  "164.312.c.2"
],
"tsc": [
  "PI1.4",
  "PI1.5",
  "CC6.1",
  "CC6.8",
  "CC7.2",
  "CC7.3"
],
"description": "File added to the system.",
"groups": [
  "ossec",
  syscheck,
  "syscheck_entry_added",
  "syscheck_file"
],
"id": "554",
"nist_800_53": [
  "SI.7"
],
"pgp13": [
  "4.11"
],
"gdpr": [
  "II_5.1.f"
]
},
"location": "syscheck",
"decoder": {
  "name": "syscheck_new_entry"
},
"id": "1779557638.3777033",
"full_log": "File 'c:\\programdata\\microsoft\\windows\\start menu\\programs\\startup\\evil.txt' added\\nMode:
realtime\\n",
"timestamp": "2026-05-23T17:33:58.778+0000"
},
"fields": {
  "syscheck.mtime_after": [
    "2026-05-23T17:33:52.000Z"
  ],
  "timestamp": [
    "2026-05-23T17:33:58.778Z"
  ]
},
"sort": [
  1779557638778
]
}

```

Wazuh FIM detection showing realtime monitoring of Startup folder persistence artifact creation

Attack Timeline

Time	Event
23:03:47	PowerShell launched
23:03:48	Startup folder artifact created
23:03:52	Sysmon Event ID 11 generated
23:03:58	Wazuh persistence detection triggered

Key Detection Findings

Finding	Impact
PowerShell-created persistence artifact	High-confidence suspicious activity
Startup folder modification	Persistence indicator
PowerShell process involvement	Increased attacker suspicion
High-value persistence path detected	Elevated investigation priority
Finding	Impact
Behavioral persistence telemetry	Improved detection confidence

Primary Investigation Artifacts

Sysmon Event IDs

Event ID	Meaning
1	Process Creation
11	File Creation

Wazuh Detection IDs

Rule ID	Severity	Description
92027	4	PowerShell process spawned PowerShell instance
92213	15	Executable file dropped in folder commonly used by malware
554	5	File added to the system (FIM/syscheck)

Investigation Outcome

The investigation successfully demonstrated how Sysmon and Wazuh provide strong visibility into Startup Folder persistence techniques initiated through PowerShell activity.

The combination of:

- PowerShell process telemetry
- persistence artifact monitoring
- Startup folder visibility
- file creation analysis
- behavioral correlation

significantly improved detection confidence and investigation quality.

Persistence was validated through Sysmon Event ID 1, which recorded automatic execution of the Startup Folder payload by explorer.exe following user logon.

The investigation also reinforced the importance of:

- persistence monitoring
- behavioral analysis
- analyst validation
- contextual investigation
- PowerShell abuse detection

during SOC operations.

Analyst Assessment

The investigation confirmed successful Startup Folder persistence establishment and execution. Sysmon Event ID 11 validated persistence artifact creation, while Sysmon Event ID 1 following user logon confirmed automatic execution of the persisted payload through explorer.exe startup processing. Correlation between Sysmon telemetry and Wazuh detections provided complete visibility into persistence creation, activation, and execution.

Conclusion

This simulation demonstrated how Sysmon and Wazuh provide strong visibility into Startup Folder persistence techniques commonly associated with malware persistence and attacker access retention activity.

Behavioral correlation between PowerShell execution and Startup folder modification significantly improved detection confidence and investigation quality.

The investigation also highlighted the importance of monitoring persistence-related filesystem locations because persistence artifacts often provide high-value detection opportunities with clearer indicators of malicious intent compared to generic network or process activity.